

CSR & Social Sector Service Data Analysis

1. Project Overview:

- **Business Analysis**

Organizations invest in **Corporate Social Responsibility (CSR)** initiatives across sectors such as education, healthcare, environment, rural development, and skill development.

The purpose of this analysis is to **understand how CSR funds and services are distributed**, identify **high-impact regions and sectors**, and highlight **gaps or under-served districts**.

- **Objective of the Analysis**

The main goals of this case study are:

- To analyze **distribution of CSR projects across districts and sectors**
- To identify **top and bottom performing districts**
- To understand **which sectors receive maximum CSR support**
- To evaluate **coverage gaps** (e.g., “Not Mentioned” districts/sectors)
- To provide **data-driven insights** for better CSR planning and decision-making

2. Data Source:

- **Source Description and Timeline:** Indian Data Portal/ CSR Projects/ Social Welfare/ 2014-2020/ Ministry of Corporate Affairs.
- **Sector :** Social Welfare/ Social Economic.

3. Problem Statement:

- ❖ **Analysis Scenarios**

The following analysis scenarios were addressed:

District-Level Analysis

- Highest CSR activity districts
- Bottom five districts by project count
- Districts with very low CSR coverage

Sector-Level Analysis

- Most funded / most active CSR sectors

- Least addressed sectors
- Sector distribution across states

State-Level Comparison

- States with maximum CSR participation
- Regional imbalance in CSR services

4. Column Details:

Column Name	Data Type	Description
ID	Integer / String	Unique identifier for each State
State_Name	String (Text)	Name of the State
State_Code	Numeric (Integer)	Code of the state
Development_Sector	Categorical	Departments
Year	Numerical	Date
District_Name	String (Text)	Name of the District
District_Code	Numeric (Integer)	Code of the District
Dev_Id	String	Indicates the Development sector id
Project_outlay_amount_in_lakhs	Numerical	It shows the value of the porject outlay
Amount_spent_in_lakhs	Numerical	It gives the value of the amount spent on the project

5. Tools & Techniques:

- **Excel**
 - Data Cleaning
 - Sorting & Filtering
 - Relationship between dimensions table and fact table.
 - Power BI for advanced visualization and querying(Charts, Bars, Tables, Slicers).
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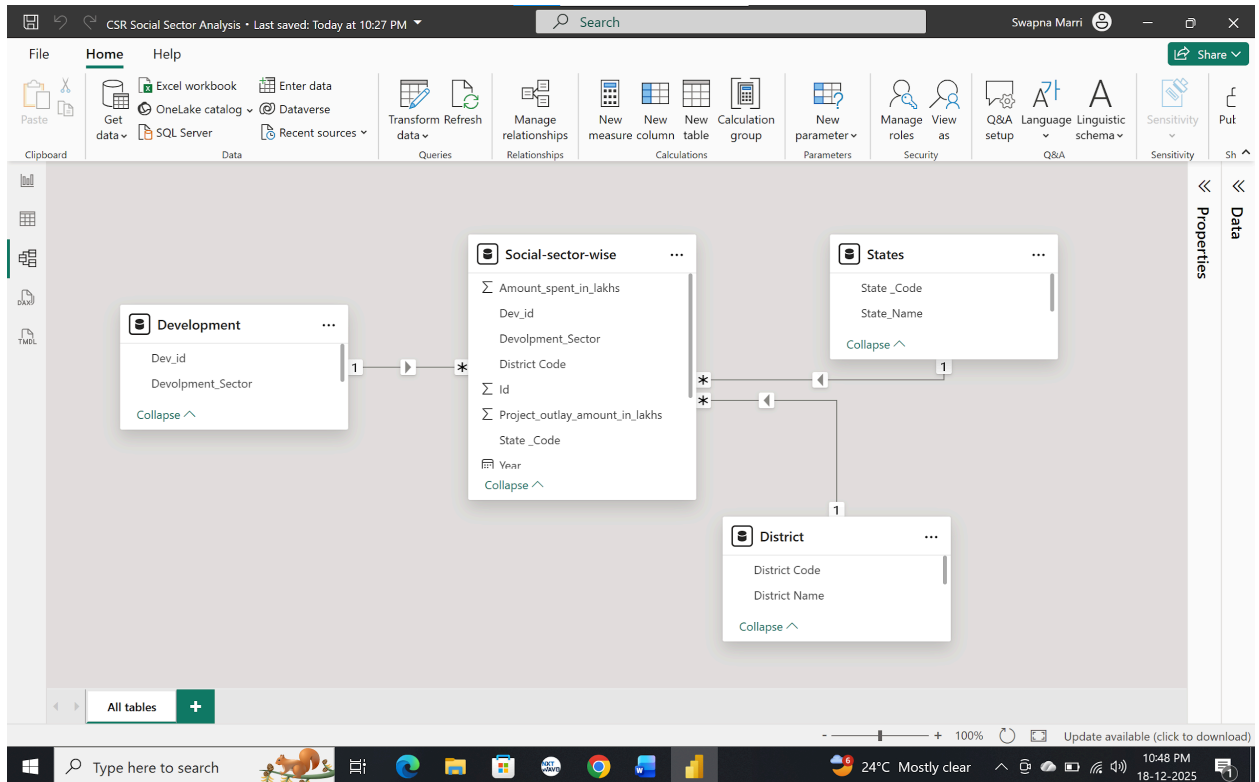
6. Data Pre-Processing (Excel):

Tasks Performed:

- **Data Cleaning & Transformation:** Removed duplicates, handled missing values, standardized formats, and created calculated fields.
- **Filtering & Sorting:** Organized data to focus on relevant records.
- Converted the data into Fact and Dimension Tables. (Star Schema)
- Removing or handling “**Not Mentioned**”
- Standardizing **district name, state names according to their respective codes.**
- Correcting **data types** (text, numbers, dates)
- Removing duplicate
- Data Modeling:

7. Data Modelling and DAX (Power BI):

- **Data Model:** Established relationships between tables, defined cardinality, its comes Star Schema.



- **Calculated Columns & DAX Measures:** Implemented DAX formulas for key metrics, such as total projects, total states, total districts, total project outlay, total amount spent on project.

Created Measure:

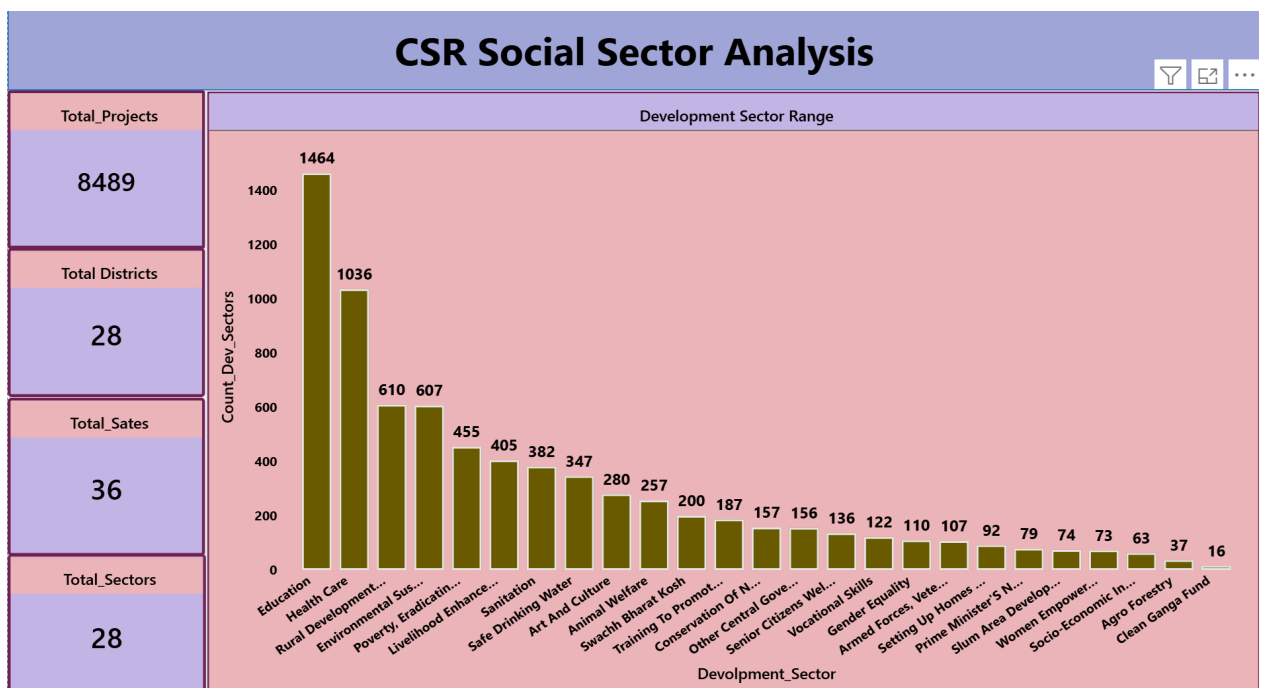
1. Depwise_Rank =

$$\text{RANKX}(\text{ALL}(\text{Development}[\text{Development_Sector}]), \text{CALCULATE}(\text{SUM}(\text{'Social-sector-wise'}[\text{Project_outlay_amount_in_lakhs}])), \text{DESC})$$
2. Total_projects = $\text{COUNT}(\text{'Social-sector-wise'}[\text{Id}])$
3. Total_sectors = $\text{COUNT}(\text{Development}[\text{Development_Sector}])$
4. Total_Districts = $\text{COUNT}(\text{Development}[\text{Dev_id}])$
5. Total_States = $\text{COUNT}(\text{States}[\text{State_Code}])$
6. Total_Project_Outlay =

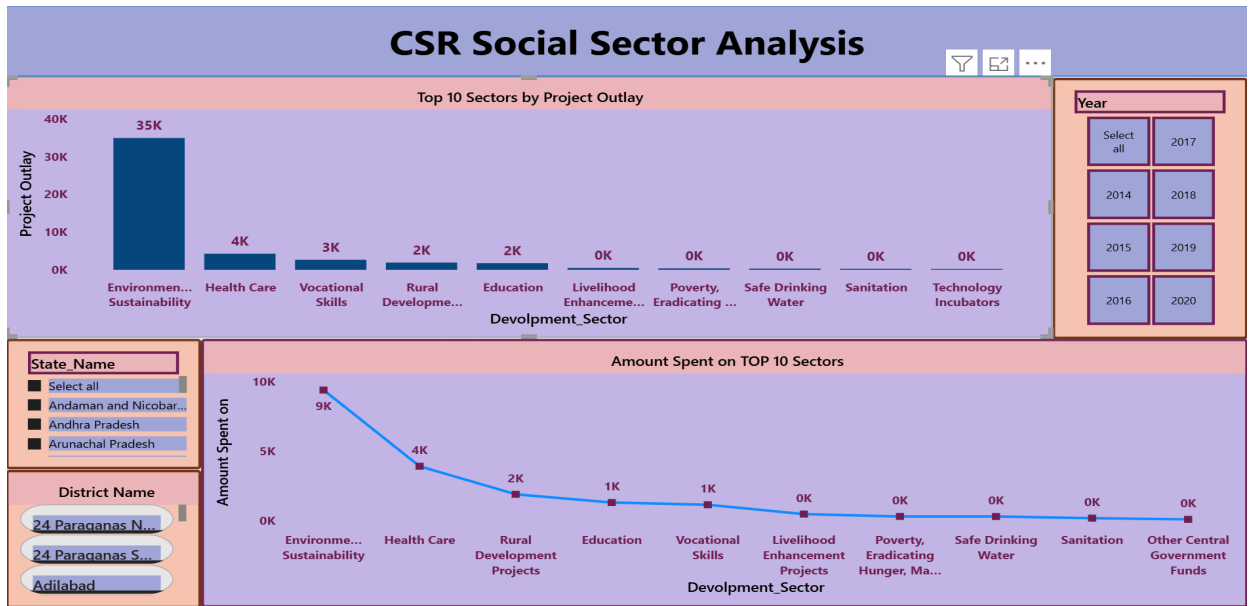
$$\text{SUM}(\text{'Social-sector-wise'}[\text{Project_outlay_amount_in_lakhs}])$$

7. Count_Dev_Sectors =
`IF(ISBLANK(COUNT(Development[Devlopment_Sector])),BLANK(),COUNT('Social-sector-wise'[Dev_id])).`
8. DAX Formula to create a Status:
 Status =
`IF('Social-sector-wise'[Project_outlay_amount_in_lakhs]>'Social-sector-wise'[Amount_spent_in_lakhs],"Low Spend",IF('Social-sector-wise'[Project_outlay_amount_in_lakhs]<'Social-sector-wise'[Amount_spent_in_lakhs],"High Spend", "Exact Spend")).`
9. Bottom 10 Projects outlays based on State Name Using a Measure:
 Bottom10_ProjectOutlay =
`CALCULATE(SUM('Social-sector-wise'[Project_outlay_amount_in_lakhs]),FILTER(VALUES(States[State_Name]),RANKX(ALL(States[State_Name]),'Social-sector-wise'[Total_Project_Outlay],,ASC)<=10)).`

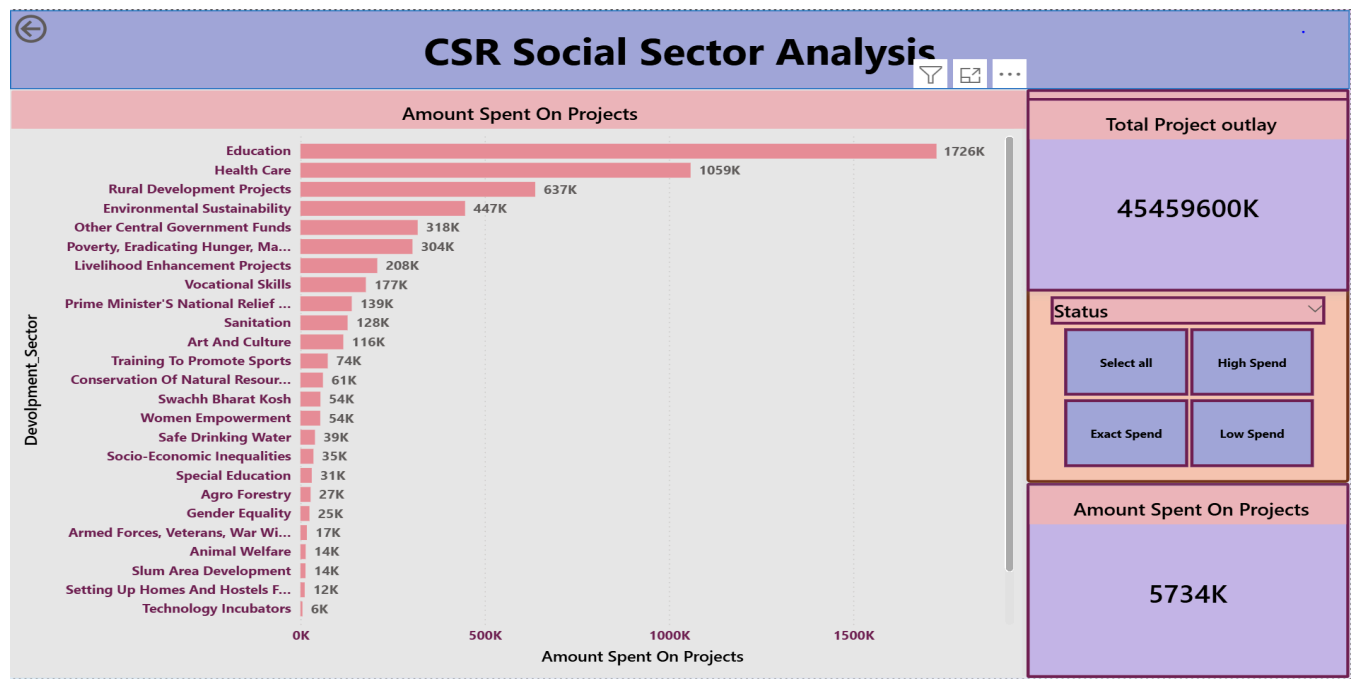
8. Analysis and Visualizations (Power BI): Analysis: Cards based on data and Development sector Analysis:



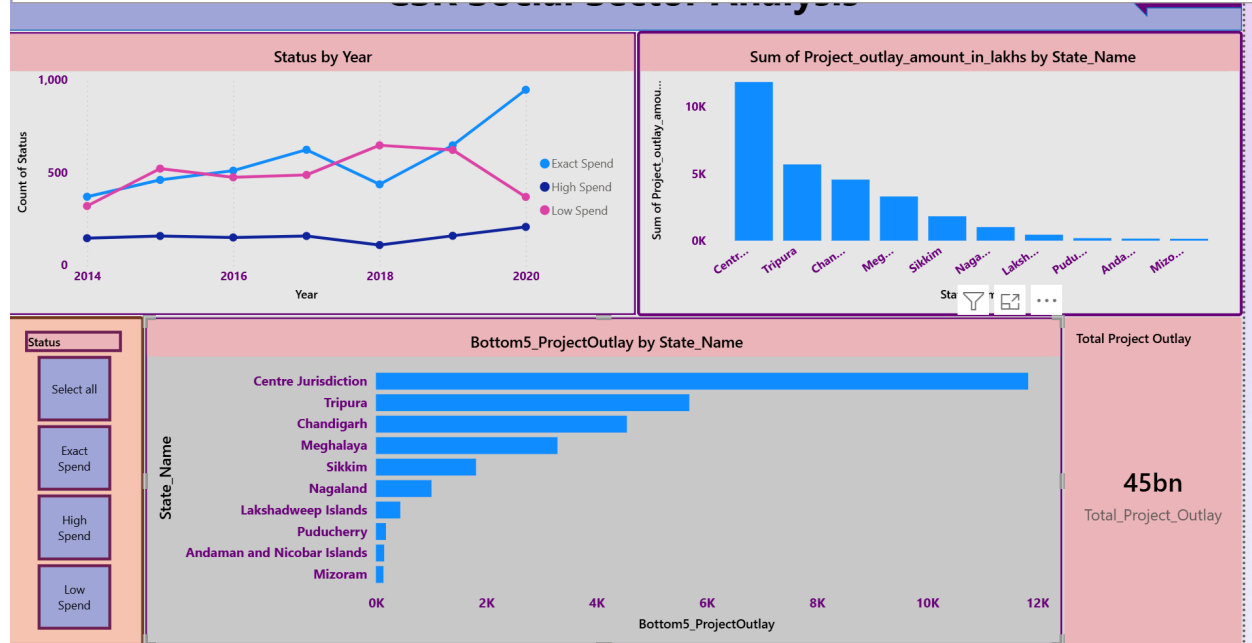
Top 10 sectors by Project Outlay And Amount spent on the Project Visuals, Slicers done for the Year, State Name, District Name. According to the slicer selection we get to know state or district Analysis.



Amount spent on projects vs Project outlay: Amount by created a conditional column based Project outlay and Amount spent Projects



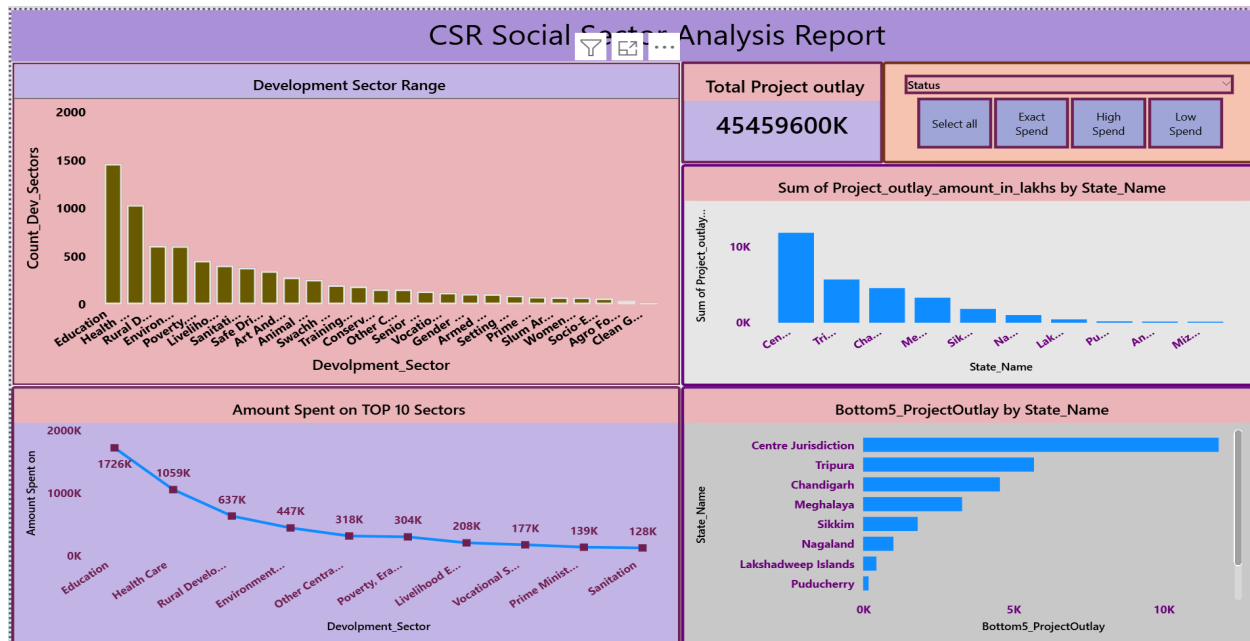
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1 Bottom5_ProjectOutlay = CALCULATE(SUM('Social-sector-wise'[Project_outlay_amount_in_lakhs]),FILTER(VALUES(States[State_Name]),RANKX(ALL(States[State_Name]),'Social-sector-wise'[Total_Project_Outlay],,ASC)<=10))
```



Matrix Created based on state name, district name, project outlay, amount spent on projects.

CSR Social Sector Analysis				
State Name VS Project Outlay VS Amount Spend				
Status	Exact Spend		High Sp	
State_Name	Sum of Project_outlay_amount_in_lakhs	Sum of Amount spent_in_lakhs	Sum of Project_outlay_amount_in_lakhs	Sum of Amount spent_in_lakhs
West Bengal	5183	5183	782	782
Uttarakhand	4715	4715	160	160
Uttar Pradesh	7003	7003	348	348
Tripura	66	66	66	66
Telangana	7280	7280	768	768
Tamil Nadu	41311	41311	1838	1838
Sikkim	407	407	78	78
Rajasthan	11486	11486	2757	2757
Punjab	2926	2926	42	42
Puducherry	118	118	6	6
Odisha	7869	7869	229	229
Nagaland	183	183	2	2
Mizoram	137	137	2	2
Meghalaya	420	420	20	20
Manipur	118	118	2	2
Maharashtra	33610	33610	25310	25310
Madhya Pradesh	6197	6197	477	477
Lakshadweep Islands	82	82	82	82

CSR Social Sector Analysis Report:



9.Insights & Conclusions:

Overall Project Distribution

- CSR projects are unevenly distributed across districts, indicating that development efforts are concentrated in a few regions.
 - A small number of districts account for the highest share of total projects, while several districts show very low or negligible participation, highlighting regional imbalance.
1. **Descriptive Insights:** CSR activities are not evenly distributed across states and sectors, with strong concentration in specific districts and domains.
 - The **total number of CSR projects** is concentrated in a limited number of districts.
 - **Top 5 States** account for a disproportionately high share of projects compared to other regions.
 - A few **development sectors dominate** the CSR landscape, indicating focused investment areas.
 - Several districts/ states show **very low** highlighting uneven regional coverage.
 2. **Diagnostic Analysis:** CSR concentration is driven by corporate proximity, ease of execution, and compliance behavior, rather than purely need-based development.
 - High-performing states and districts are likely **industrial** attracting more CSR initiatives.
 - Low-performing districts may suffer from:

1. Lack of corporate presence
 2. Limited awareness of CSR programs
 3. Weak implementation capacity
- Sector dominance suggests companies prefer high-visibility, low-risk sectors (**e.g., education, healthcare**).
3. **Predictive Analysis:** Without strategic intervention, CSR initiatives will become more concentrated, increasing regional and sectoral inequality.
 - Top districts will continue receiving the majority of CSR projects.
 - Low-performing districts will remain underserved.
 - Dominant sectors will continue to attract **most CSR funding**.
 4. **Prescriptive Analysis:** Strategic reallocation, improved data governance, and impact-driven metrics can make CSR efforts more equitable, transparent, and effective.
 - **Redistribute CSR focus** toward underserved and low-performing districts.
 - Introduce **minimum State-level CSR allocation guidelines**.
 - Encourage investment in **underrepresented development sectors**.
 - Improve data quality by:
 - Standardizing district and development sectors.

10. Conclusions:

CSR activities appear to be more **compliance-driven than impact-driven** in some areas. There is a strong opportunity to:

- Expand CSR reach to underserved districts
- Align CSR projects with **regional development needs**
- Improve **equitable distribution of funds**

While CSR initiatives show strong activity in select districts and sectors, targeted policy intervention and data-driven planning are essential to ensure equitable and impact-focused development.